

Chapter 1

Conclusion

This final chapter reviews the work presented in this thesis. It begins by revisiting each research question in light of our contributions. It then discusses the limitations of these contributions. Next, it presents the directions for future research that these limitations open. Finally, it closes with a broader reflection on this work.

1.1 Summary of Contributions

This section revisits the four research questions introduced in Section ?? . For each of them, it summarizes the answer provided by this thesis.

Research Question ?? concerned the approaches proposed in the literature to analyze the internal workings of Neural Networks (NNs). To answer it, we first made explicit the working hypothesis shared by most of this literature, the *Concept Hypothesis* (Hypothesis ??). According to this hypothesis, NNs generalize by organizing their Latent Spaces (LSs) around concepts. We then organized existing methods into four families: probing, interpretable directions, Sparse Autoencoders (SAEs), and clustering.

Research Question ?? concerned the organization of existing Explainable Reinforcement Learning (XRL) methods [1, 2]. Our starting point was to clarify what an explanation means in the context of Reinforcement Learning (RL). Building on this framework, we identified the notions of subject and source of an explanation. Regarding the subject, we distinguish two levels: a single decision and the agent’s strategy. For the source, we identify three possibilities: policies, trajectories, and value functions. From these elements, we proposed a taxonomy that accounts for the specificities of RL and used it to review the state of the art.

Research Question ?? concerned the use of Latent Representations (LRs) to identify the mechanisms underlying the decisions of RL agents. To address it, we introduced Clustering Agent Representation (CAR), an XRL method based on the principles of Mechanistic Explainability (ME) [3]. CAR trains several surrogate policies independently to reproduce the same agent. It then clusters their LRs and compares the resulting partitions. This procedure relies on the *Representation Convergence Hypothesis* (Hypothesis ??), according to which networks trained on the same task develop similar concepts. Applied to six

Atari environments, CAR recovers similar clustering structures across surrogate policies. The qualitative analysis shows that these clusters capture interpretable aspects of the task.

Research Question ?? concerned the objective quantification of explanation quality. We addressed it by introducing three metrics. The *Clustering Universality* (CU) measures the consistency of the clusters across surrogate policies. The *Abstraction Score* (AS) measures how much these clusters differ from the structure of the observations, the actions, and an untrained network. The *Adjusted Clustering Universality* (ACU) combines both properties into a single measure and provides a practical criterion for selecting the layer to analyze.

Throughout these contributions, we developed three hypotheses that form a coherent chain. The *Mechanistic Hypothesis* (Hypothesis ??) states that NNs can be understood through their internal mechanisms. The *Concept Hypothesis* (Hypothesis ??) states that these mechanisms are organized around concepts. The *Representation Convergence Hypothesis* (Hypothesis ??) states that these concepts reflect the task. CAR translates these hypotheses into an empirical procedure, and the experimental results provide support for this framework.

1.2 Limitations

This section discusses the four main limitations identified at this stage.

First, CAR describes how the LS of an agent is organized, but not what each cluster means. The CU, AS, and ACU metrics quantify the consistency and abstraction of the clusters, but they say nothing about their content. In this thesis, the meaning of each cluster was assigned through manual inspection of the corresponding observations. This step is time-consuming, subjective, and difficult to scale to a large number of clusters or environments.

Second, the clusters identified by CAR are not yet directly related to the decisions of the agent. We observe that the agent distinguishes certain situations, but not how these distinctions lead to a given action. As a result, CAR does not explicitly explain the behavior of the agent.

Third, the experiments were conducted on six Atari environments. Whether CAR scales to more complex environments and larger networks therefore remains an open question. In addition, only one clustering method was evaluated, based on k -means with Euclidean distance. Other distances and other clustering methods could reveal different structures in the LS. Since the pipeline of CAR is modular, these alternatives can be tested and compared using the proposed metrics.

Finally, these metrics have only been applied to clustering. The other families of methods reviewed in Chapter ?? have not yet been evaluated within this framework. This is particularly the case for methods based on the *Superposition Hypothesis* (Hypothesis ??). These methods are widely used, but their assumptions remain debated. Our metrics could be used to test these assumptions quantitatively, and to compare these methods with CAR on the same tasks.

1.3 Perspectives

The limitations discussed above suggest several directions for future research. This section presents four of them. For each of them, we first identify the main scientific challenge it raises. We then describe the approach we propose and how it could be validated.

1.3.1 Automatic Textual Description of Clusters

As discussed, the semantics of the clusters is currently assigned manually. The challenge is to obtain a reliable textual description of each cluster automatically.

One possible approach is to first extract, for each cluster, the video segments formed by consecutive observations belonging to it. Each segment is then described in natural language by a vision-language model. Finally, the descriptions of all segments of a cluster are synthesized into a single description of the cluster.

The reliability of these descriptions can be assessed through a classification test. Given a new video segment, not used to produce the descriptions, a model must identify the cluster to which it belongs, using only the textual descriptions of the clusters. The predicted cluster is then compared with the one assigned by CAR. If the classification accuracy is high, the textual description faithfully captures what distinguishes the cluster. This would fully automate the CAR method, from the identification of the clusters to their interpretation.

1.3.2 From Latent Similarity to the Strategy of the Agent

The goal here is to derive an explanation of the strategy of the agent from the similarity of its LRs. The main challenge is that the clusters group states that the agent perceives as similar, but provide no information about the dynamics of its interaction with the environment. They must therefore be turned from a static view of the LS into a dynamic view of the behavior of the agent.

Each cluster can be treated as an abstract state. The transitions observed between clusters during episodes then define a simplified Markov Decision Process (MDP). This MDP can be represented as a high-level graph. Its nodes are the concepts identified by CAR, and its edges describe how the agent moves between them. This graph thus summarizes the strategy of the agent.

Such a graph is useful only if it is truly informative for an observer. An observer with the graph should be able to describe the behavior of the agent better than an observer without it.

1.3.3 Human Evaluation of Explanations

The objective is to verify that the explanations produced by CAR actually improve the observer's mental model of the agent. This would also confirm the relevance of the associated metrics, namely CU, AS, and ACU. The main challenge is that understanding has no consensual operational definition, which makes it difficult to measure.

Such an evaluation requires an experimental protocol involving human observers. This protocol should be defined in collaboration with researchers in cognitive science and psychology. It would then be tested by comparing the understanding of observers with and without the explanations of CAR.

120 1.3.4 Hypotheses on the Structure of the Latent Space

121 CAR assumes that concepts are organized as clusters, whereas the *Superposition*
 122 *Hypothesis* (Hypothesis ??) assumes that they are encoded as directions. The
 123 objective is to determine which of these hypotheses best describes the LS of a RL
 124 agent. The main challenge is that the two families of methods do not produce
 125 the same kind of object, a partition in one case and a set of directions in the
 126 other, which makes them difficult to compare.

127 The metrics proposed in this thesis can serve as a common framework. They
 128 can be applied to the concepts extracted by methods based on superposition,
 129 provided that these concepts are expressed as a partition of the observations.

130 Both hypotheses can then be evaluated with these metrics on the same agents
 131 and environments. The hypothesis that yields the most universal and abstract
 132 concepts would be the one that best describes the structure of the LS.

133 1.4 Final Remarks

134 To conclude, let us step back and consider this work as a whole. We propose
 135 explainability methods that build on the specific properties of NNs. The goal is
 136 to uncover how these networks organize information, in order to shed light on
 137 their internal functioning. From this perspective, explainability is not achieved
 138 despite the use of NNs, but through them. The results presented in this thesis
 139 show that this approach is feasible. It thus offers a way to better understand
 140 our creations and to ensure that they remain aligned with our intentions.